Economic Geography

ISSN: 0013-0095 (Print) 1944-8287 (Online) Journal homepage: https://www.tandfonline.com/loi/recg20

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To cite this article: Roberto Camagni, Roberta Capello, Silvia Cerisola & Ugo Fratesi (2020): Fighting Gravity: Institutional Changes and Regional Disparities in the EU, Economic Geography, DOI: 10.1080/00130095.2020.1717943

To link to this article: https://doi.org/10.1080/00130095.2020.1717943

Published online: 14 Apr 2020.

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Fighting Gravity: Institutional Changes and Regional Disparities in the EU

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abstract

A thorough investigation of the relationship between the deepening and the widening processes of EU integration and the historic evolution of regional disparities is missing in the related vast existing inductive literature. This usually focuses on EU15 countries or Central and Eastern European Countries, generally involves relatively short and recent periods, and takes into consideration just one institutional change at a time, if any. This article aims at filling these gaps by providing a theoretical interpretative framework on the effects of each institutional change on inter- and intranational disparities, and by verifying the fitness of the theoretical expectations with a longitudinal trend analysis and with an econometric event analysis. Theory was overall right in claiming that widening and deepening of the EU would have exacerbated intraregional disparities. Counteracting such tendency could look like *fighting gravity*, since disparities continuously reappear in different forms and spatial levels, transmitting a sense of unescapable normality.

ECONOMIC GEOGRAPHY 1 00(00):1–29. © 2020 Clark University. www.economicgeography.org

Key words:
disparities
institutional changes
EU enlargement
single market
common currency
Euro
Theil index

2 Acknowledgments

The descriptive empirical part of this work is based on a research project carried out by the authors for the European Commission, Directorate General for Regional and Urban Policy [Contract No 2017CE16BAT125].

After more than sixty years from the Treaty of Rome, the mere existence of the EU is challenged as never before. The current political and economic instability within the EU calls for further analysis to shed light on the origins and causes of a situation that led—among other consequences—to the rise of populist and nationalist parties all over Europe (Dijkstra, Poelman, and Rodríguez-Pose 2019). The results of the recent European elections are indeed a clear signal in this direction, especially in countries, such as France, Italy, and the UK, that seem to have lost or forgotten the benefits accruing from EU membership (Eichengreen 2007; Eichengreen and Boltho 2010; Schout and Bevacqua 2018; Campos, Coricelli, and Moretti 2019).

Among these benefits, a long-run process of convergence that has historically characterized the European countries and regions seems to have halted in recent years. Some world macroeconomic elements, and especially the recent economic crisis, have in fact stopped the (international) convergence that had characterized the evolution of the EU. Increasing intranational disparities and nondecreasing international disparities among EU member countries (at least since the outburst of the Great Recession) could be indeed a driver of discontent and—along with persistently low levels of economic growth and lack of opportunities in some areas—a threat to social cohesion and political stability (Rodríguez-Pose 2018).

The relationship between European integration and regional inequalities is not clear-cut from a theoretical point of view. On the one hand stands the *neoclassical* thought that trade integration is a long-term process that eventually leads to a decrease in disparities. This would occur because lagging areas would grow faster than others given their cost advantage (Solow 1956; Swan 1956) and because of factor-price equalization due to specialization in low labor-intensive sectors (Ohlin 1933; Heckscher 1991).[1] This last process is emphasized by production factors increasing international mobility, which is equal to zero in the assumptions of the standard international trade theory (Samuelson 1949; Borjas 1989). On the other hand, is the idea that the benefits and the costs of integration

[1] The title itself of Ohlin's (1933) book underlines the profound difference between international and interregional trade, suggesting that in the latter case, equalization could depend mainly on out-migration processes from lagging regions rather than on pure development.

are unlikely to be shared equally. In this case, more advanced regions (usually urban or capital regions) are expected to benefit more, while more disadvantaged ones (often in the periphery) are at risk of falling behind, due to specialization in traditional, slow-growing sectors. In this case, economic integration would lead to agglomeration of production and concentration of income. This approach draws on the *cumulative causation* approaches (Myrdal 1957; Hirschmann 1958), *growth poles* theory (Perroux 1955), the *endogenous growth* models that endogenize technological change into investment processes (Romer 1986; Lucas 1988; Barro 1990), and on the New Economic Geography (NEG) contributions (Krugman 1991; Fujita 1993; Venables 1996).

In order to interpret regional disparities, both theory and evidence favor the distinction between inter- and intraregional disparities. In fact, on theoretical grounds, the condition of countries and regions in the integration process is very different, since the latter do not dispose of powerful policy tools, such as control on exchange rates and money supply, and operate in the same integrated labor market with advanced areas sharing similar wage levels (Camagni 2002). On empirical grounds, despite the initial optimistic belief that greater and greater integration would have led to a process of continuous convergence, this has actually occurred in the EU only for international disparities, which have been steadily declining until the recent economic crisis. Being European integration is a process in favor of a more dynamic, competitive, and modern economy (Giannetti 2002; Heidenreich and Wunder 2008), intranational disparities were instead expected to increase, making the adoption of *regional and cohesion policies* to limit inequality a necessity (Camagni and Cappellin 1985; Dunford 1994; Camagni 1995).

Even if the need to reduce disparities in terms of development levels across the different EU regions is directly related to some of the basic principles of the EU (European Union 1986; Ezcurra and Pascual 2007; Ezcurra, Pascual, and Rapún 2007a; Ezcurra and Rodríguez-Pose 2009; Petrakos, Kallioras, and Anagnostou 2011; Alcidi 2019),[2] such endeavor could become a matter of *fighting gravity*, since deeper European integration entails natural pushes favoring more advanced regions. This happens because more advantaged areas tend to be the main beneficiaries of new investments and to be better prepared to compete on the international market, while weaker areas might lack some basic preconditions for modern development. In addition, the desired long-term decrease in regional disparities could become a matter of *fighting history*, that is, challenging long-term cultural, social, and economic differences.

Having all these controversial views in mind, it emerges quite clearly how most existing studies are mainly inductive, that is, they tend to narrate rather than test. In addition, they usually focus on either EU15 countries or Central and Eastern European Countries (CEECs), and usually involve relatively short and/or recent periods, while it has been demonstrated that picking different economic phases brings different results for what concerns convergence/divergence. Last, but not least, in most works the distinction between inter- and intraregional disparities is either not explicitly clarified or not taken into consideration.

[2] Article 130a of the Single European Act (1986) states that "[I]n order to promote its overall harmonious development, the Community shall develop and pursue its actions leading to the strengthening of its economic and social cohesion." Single European Act, February 17, 1986, 1987 O.J. (L 169) 1, 25 I.L.M. 506. The Treaty of Rome (1957) itself was mentioning the goal of a harmonious development of economic activities (art. 2) *"en réduisant l'écart entre les différentes régions et le retard des moins favorisés"* (Preambule) [by reducing the gap between the different regions and the backwardness of the least advantaged (Preamble)]. EEC Treaty or Treaty of Rome: Treaty Establishing the European Economic Community, March 25, 1957, 298 U.N.T.S. 3, 4 Eur. Y.B. 412, https://eur-lex.europa.eu/legal-content/FR/TXT/PDF/?uri=CELEX:11957E/TXT&from=IT.

This article aims at putting some order in the vast existing literature on the topic and at overcoming its shortcuts. In particular, it aims to provide an interpretative theoretical framework summarizing the expectations on the role of radical institutional changes—both the agreements leading to the widening and deepening of the EU—on regional disparities; not mentioning the initial foundation of the EC itself, these institutional changes happened with the widening of the EU at each enlargement,[3] at the deepening of the EU, through the establishment of the *single market* (1993, but even earlier, at the commitment to it of President Delors in 1985), and at the introduction of a *common currency* (1999) (see "Theoretical Expectations on the Effects of Institutional Changes on Regional Disparities").

The theoretical expectations are then verified empirically, by exploiting NUTS2-level data on the level of gross domestic product (GDP) per capita for a long time span (from 1980 to 2015) for twenty-eight EU countries (see "Methodological Approach"). The link between relevant institutional changes within the European integration process and the trend in disparities is explored at both the inter- and the intranational levels, in order to detect specific and potentially diverging behaviors at the two different spatial levels (see, respectively, "The Evolution of Intercountry Disparities in the EU: A Long Run Convergence" and "Fighting Gravity: Long Run Intracountry Divergence"). To achieve its aim, the study calls for a synchronization of the entry time in the EU. This step allows exploration of the way in which the integration affected disparities and wealth in the new members (see "Synchronizing Entry Time in the EU: Intracountry Disparities and Stages of Development").[4] The main results stemming from the longitudinal trend analysis are verified by an econometric event analysis (See "Causal Link between Institutional Changes and Regional Disparities: An Event Analysis Approach"), and main messages confirmed, namely, that the theory was right in claiming that the enlargements exacerbated intranational disparities, where richest areas were the ones gaining the most from the integration process. The theory was moreover right in foreseeing intranational divergence due to the establishment of the single market. Instead, the theoretical expectations on the increase of intraregional disparities due to the introduction of the common currency did not find evidence, probably as the result of the effectiveness of the counteracting interventions. Concluding remarks are presented in "Conclusions."

Theoretical Expectations on the Effects of Institutional Changes on Regional Disparities

The Common Market and Regional Disparities

The advantages of the European integration process were widely analyzed, discussed, and empirically evaluated by the scientific *milieu* in parallel with each of the main phases of the integration process itself, often anticipating the formal institutional decisions and supporting their subsequent implementation. Institutional changes are in fact prepared and determined with clear time schedules. Thus, the economic systems can prepare for such changes.[5]

[3] Given data availability, the empirical analysis starts in 1980. We are therefore forced to analyze the enlargements from that of 1981, leaving aside the 1973 one. Theoretical considerations presented in "The Common Market and Regional Disparities," however, hold also for the six founding members and for the 1973 enlargement.

[4] On the synchronization of *business cycles*, the reader can refer to Petrakos, Rodríguez-Pose, and Rovolis (2005).

[5] The years leading up to official membership status constitute what can be defined as *enlargement shock* (Gomulka 2003; Charron 2013).

This section aims at recalling the above-mentioned debate by summarizing the theoretical reasons behind the expected economic advantages of the different institutional changes and by highlighting the spatial distribution of such advantages.

The themes, and consequently the theories on which the debate was focusing, are classified according to the main institutional phases of the European integration, both in terms of widening and deepening of the EU: economic integration starting with the Common Market and its first enlargements, the single internal market, the monetary union, and the 2004 enlargement to CEECs (Table 1).

The creation of the Common Market, and the subsequent economic integration phases, generated a large debate over its potential economic effects, which were expected to be positive.[6] The major issue concerned the welfare and growth effect of the creation of the European trade union, with the abatement of internal tariffs among the six initial partners.[7] The main theoretical tools utilized were the classical and neoclassical pure theory of international trade, investigating the gains from trade (Södersten 1971). Both the classical Ricardian principle of comparative advantage and the neoclassical Heckscher-Ohlin model of trade based on differences in availability and price of production factors among countries grant an advantage for all trade partners (irrespective of their initial relative efficiency) in opening trade and in the consequent specialization and international division of labor. The main advantages that were expected to occur came from the impact of integration on static efficiency and, namely, on the allocation of resources within the union and the rest of the world (Sapir 2011). In line with the theorem of factor prize equalization, the integration effect was expected to reduce differences in factor prices within a custom union through the mobility of capital, labor, and entrepreneurial resources, and their more efficient use (Balassa 1961; Sapir 2011).

The optimistic neoclassical view was reinforced by the idea that dynamic advantages could be generated by integration, giving emphasis on issues of market size and scale economies (Viner 1953): in particular (1) a marked increase in competition on markets; (2) greater economies of scale in the production of goods because of the larger size of outlet markets; (3) the creation of trade in final and intermediate goods; (4) increased investments prompted by forecasts of greater competition; (5) an expected demand for a greater variety of goods (income effect); (6) a shift in demand for goods produced with more efficient techniques, guaranteeing lower prices for the same quality; and (7) technology and knowledge transfers from strong to weak regions (Viner 1953; Scitovsky 1958; Lipsey 1960).

Theories did not deny the possibility of negative effects from integration. In fact, the custom union theory proposed the possibility for a trade diversion—when a country refuses to trade with an external partner, which produces at lower prices, and redirects its imports toward internal partners, that produce at higher prices—limiting the positive effects of a trade creation among partner countries. Moreover, when the effects of the engagement to free trade on internal income distribution—which are not considered by the main theories—are taken into account, the picture might not be so optimistic. In fact, social classes hit by the structural transformation of the economy—by strong jobs reductions in the abandoned or shrinking sectors—should be compensated by the benefited classes through redistributive policies, reducing therefore the hypothesized aggregate advantage for countries and sectors (Södersten 1971). Another strong

[6] We refer here to the establishment of the Common Market (six members), and the subsequent enlargements to Denmark, UK, and Ireland in 1973 (EEC9), to Greece in 1981 (EEC10), to Spain and Portugal in 1986 (EEC12), and to Austria, Finland, and Sweden in 1995 (EU15).

[7] The reduction of transaction costs came along with a reduction of uncertainty.

Table 1

Institutional Changes: Theoretical Expectations on Economic Effects and on Regional Disparities

Institutional Changes	Expected Economic Effects	Expected Effects on Intercountry Disparities	Expected Effects on Intracountry Disparities
Economic integration (first four enlargements since the Common Market)	Increase in trade, market potential, economies of scale, knowledge, and technology diffusion	Convergence through - market potential advantages especially in presence of favorable conditions in weaker country - labor cost advantage for less advanced countries, technological catching up - anticipatory effects on investments - synchronization of regional business cycles	Short-term divergence - weak competitive capacity in weaker areas of the countries - lack of some crucial preconditions for growth Convergence in longer term, with selective diffusion through geographic proximity and toward intermediate income regions, slowed down by permanent weakness and negative trade balance in lagging areas
Single market	4 free movements (goods, services, capital, labor), removal of nontariff barriers, labor market integration, recognition of education degrees	Convergence through - enhanced and enlarged integration effects - driving effects from external investments from advanced non-European countries but also from advanced European countries, particularly in weaker countries	Divergence, due to - more advanced urban areas hosting widest share of investments in services (finance, insurance and commerce)
Common currency	Common European monetary policy Countries renouncing to national monetary policies (devaluation, issue currency) Lowering of interest rates, especially in countries with previous weaker currencies	Convergence through - new trade opportunities (lower transaction costs especially in weaker countries) - *internal devaluation* through labor cost decrease in weaker countries - lower interests on debt paid by countries with high public debt Limited by - recessionary fiscal adjustments required for weaker countries	Potential divergence limited by - enhanced and widened regional development policies - industrial areas of weak countries penalized by the lack of devaluation policies
2004 and subsequent enlargements to Eastern countries	Greater availability of low-cost labor for the old member countries Integration advantages for the new member countries (NMCs)	Convergence through - labor cost competitiveness of NMCs - diffusion of integration advantages particularly FDI and technology inflows - geographic centrality of Germany and positive balance of advantages for NMCs Limited by - decrease of EU funds to old weak countries in favor of new entrants	Divergence through - greater integration advantages for advanced (urban) regions and regions closer to the EU border in new member countries - decrease of EU funds to weak regions in old member countries

assumption in both the classical and neoclassical theory of international trade concerns the condition of full employment: workers displaced by the transformation do not suffer unemployment but only a lower salary in the new employment sectors, for which they do not have the necessary skills (see Table 1).

The ex ante expectations about the quantitative outcomes of the launching of the European Common Market were surprisingly not huge at all, if not "insignificant" (Scitovsky 1958, 67); according to Verdoorn (1954), there was lower than a 1 percent of GDP increase. This was a consequence of the particular approaches utilized in the analyses, mainly comparative static.[8] In fact, when taking into consideration dynamic effects, such as the presence of unutilized economies of scale in industries or the effects of the abrupt change in competitive climate, the result is different (Scitovsky 1958; Södersten 1971).

The spatial distribution of the expected positive effects of market integration is in favor of a convergence trend of GDP per capita among countries. The enlargement of the market potential, the increase in trade, the increase in economies of scale, and the diffusion of technological and managerial knowledge are especially of greater advantage to less developed countries whose less advanced industrial structure benefits from managerial, organizational, and technological catching-up processes. More importantly, less developed countries enter the large integrated market with labor cost advantages, becoming the most favorable locations for new capital accumulation and for new cumulative causation processes à la Myrdal (1957). Finally, integration was expected to also progressively increase the synchronization of business cycles at the national and regional levels (Barrios and De Lucio 2003), although this was already present from the earlier years of the union (Christodoulakis, Dimelis, and Kollintzas 1995).

The impact of the economic integration on within-country disparities was instead not so straightforward to theorize, and initially it was hardly taken into account. At least in the short term, divergence was expected. Starting from an interesting passage by Tibor Scitovsky written as early as 1958, an expectation emerged about a likely redistribution of production among sectors and toward more efficient firms inside each sector—consequent to trade liberalization—favoring more advanced and efficient areas. International specialization and trade creation (Scitovsky 1958) would generate the crucial need and request for "a pan-European capital market and a common pan-European employment policy" (Scitovsky 1958, 95), in the form of a "planned and publicly controlled" regional development and investment policy (Scitovsky 1958, 134).[9]

[8] Other possible reasons for this low impact, assessed both ex ante and ex post, were the fact that, according to the comparative advantage theory, the advantages of international specialization do not reside in the differences between production costs of the single countries, but rather in their variation around the international (necessarily lower) average (Scitovsky 1958); and the fact that most of the new trade among European countries happened at the intraindustry level (or *two-way* trade), where the advantage mainly refers to variety of available goods rather than to their cost (Balassa 1975).

[9] These would be, according to the title of chapters, "The Problems of Currency Union" (Ch. II.B) and "Some Likely Consequences of Integration" (Ch. III.4) (Scitovsky 1958). These interregional effects of integration are likely to be faster and wider than the international ones. They can also generate demographic abandonment and even desertification of some internal or peripheral areas, due to easier interregional mobility of labor and absence of reequilibrating automatic mechanisms in the economic sphere such as currency devaluation and downward flexibility of prices and wages—existing at the international level (Camagni 2002). Empirical evidence in the EU confirms rapid increases in the early phases of integration (Italy in 1955–65; Spain and Portugal in the 1990s, New Eastern accession countries in the 2000s), correlated with phases of acceleration of the integration process, like in the years 1985–95 (Camagni and Capello 2014).

Other arguments were presented concerning the difficulties facing relatively backward regions in the context of European trade integration, namely:

- The *cumulative causation* argument (Myrdal 1957) indicating a likely virtuous cycle in stronger regions—generated by in-migration, creation of new local demand, new investments, consequent new growth—and an opposite vicious cycle in weak, out-migration regions. A similar process is due to take place at the urban/metropolitan scale as a consequence of the (cumulative) agglomeration economies effect;
- The *double challenge* argument referring to the uncomfortable condition of weaker regions in advanced countries, challenged both by the more advanced regions and by emerging countries (Holland 1977);
- The conceptual dissymmetry between integration problems at the international level and those concerning interregional relationships at the intranational level. Different than countries, regions do not control exchange rates or money supply, and their internal wages, being defined by national bargaining processes, are not sufficiently flexible. In case of a lower productivity (or a lower productivity increase over time) with respect to stronger regions, their fate is unemployment and out-migration (something that happened very often in remote or internal areas, with wide disadvantages) (Camagni 1995, 2002).[10]

In addition, in open, transnational spaces some (more advanced) regions can be in a better position to compete, placing themselves successfully as regional knots in global service, innovation, and production networks, while other (less developed) regions in the same country may fail in this respect (Heidenreich and Wunder 2008; Petrakos, Kallioras, and Anagnostou 2011).

Progressive convergence in productivity was instead expected in the long term. Also in this case, however, convergence was conceived only when the diffusion of growth through geographic proximity was able to overcome the negative trade balance registered by weak regions, representing the natural markets for the more integrated economies of stronger regions.

The Single Market and Regional Disparities

The second phase of European integration, encompassing the last two decades of last century, was launched by President Delor's White Paper on the Completion of the Internal Market approved at the Council of European Heads of State and Government in Milan (June 1985) and by the following Single European Act approved at the Luxembourg Council (December 1985). The two documents presented the new strategy for an effective abolition of internal frontiers, the legislative reforms that were crucial for the approval of the connected new rules and regulations (about 300)—namely, Parliament approval at a qualified majority—and the financial means required for the creation by 1992 of a Great Single European Market of 320 million people, along with its accompanying measures. The targets were the still persistent nontariff obstacles to trade; frontier and customs controls; barriers to movement of people, services, goods, and capital; mutual recognition of diplomas and intellectual property

[10] In economic terms, regions do not compete on the basis of a Ricardian, *comparative advantage* principle, but rather on a Smithian, *absolute advantage* principle (Camagni 2002). The Ricardian principle, guaranteeing each country a role in the international division of labor, whatever its relative efficiency or competitiveness, is based on two alternative adjusting mechanisms—wage-price flexibility or currency devaluation—which are not available at the regional level.

rules; and opening of public procurement. A widening of the regional development funds was clearly and openly indicated as necessary for avoiding the new regional disparities that could follow from the new integration rush and for a renewed social and economic cohesion policy (Dunford 1994; Petrakos Kallioras, and Anagnostou 2011).

Convergence in productivity levels was again foreseen. All the advantages expected to take place with the establishment of the Common Market for goods were expected to be reinforced thanks to the creation of the single market encompassing goods, services, people, and capital, enhancing convergence between countries, with the weaker nations being able to gain the most. These effects were clearly anticipated, especially by large multinational enterprises, starting in 1985, the year of the official launching of the single market project. A long period of investment expansion happened, ending significantly in 1993 (the year of the official start of the reform). Multinational enterprises were at the core of this process. EU countries in 1991–92, in fact, received up to 50 percent of world foreign direct investment (FDI) (UNCTAD 1997; Camagni 2001), a share never attained before (nor later). The investments involved mainly tertiary sectors and were predominantly directed to the *gateways* of the internationalization of these countries, namely, the largest cities, boosting a land rents bubble in these cities exactly in the same period, 1985–92, and in particular the rental value of office spaces, outweighing the trend in medium and small cities (Camagni and Gibelli 1996). As a result, divergence was the main expectation when intranational disparities were taken into consideration. More advanced urban areas—also characterized by a greater concentration of highly qualified people—were those capitalizing the advantage of the single market, enlarging the gap with small and medium city regions and rural areas (Dunford 1994; Capello 2002).

The Common Currency and Regional Disparities

Another important step forward in the integration process coincides with the *establishment of the monetary union and the common currency, the euro*. Of course the analyses of its potential effects and the scientific debate started much earlier, at least since the seminal article of Mundell (1961), the Maastricht Treaty (1992)[11] stating the roadmap to it, and the consequent scientific contributions (De Grauwe 1994; Baldwin 2006; Baldwin et al. 2008; Eichengreen and Boltho 2010).

A thorough analysis of the different attitudes and positions goes far beyond the aims of this section. Suffice here to recall the fields in which the costs and advantages of the monetary union are expected. "Whereas the costs of a common currency have much to do with the macroeconomic management of the economy," namely, with the costs for the single countries of relinquishing a powerful tool of economic policy such as monetary policy, "the benefits are mostly situated at the microeconomic level" (De Grauwe 1994, 60). With regard to the latter, they are found in the elimination of transaction costs on currencies (estimated from 0.5 percent to 1 percent of GDP); the reduction of incentives for firms to international price discriminations for the same products (with wider benefits but more difficult to measure); the reduction of uncertainty for both firms and consumers (with expected positive effects at least for risk-averse actors); and, above all, in the much more efficient working of the price mechanism in the allocation of resources (De Grauwe 1994).

Costs—and in particular costs for renouncing macroeconomic adjustment tools (and the macroeconomic balance between costs and benefits)[12]—depend (inversely) on openness to

[11] Maastricht Treaty, TEU or Union Treaty: Treaty on European Union, February 7, 1992, 1992 O.J. (C191) 1, 31 I.L.M. 253.

[12] For a debate on the costs of the common currency, see Wyplosz, Nickell, and Wolf (2006).

international trade of the single countries, on their homogeneity in macroeconomic fundamentals and conduct of economic policy, on the individual and collective acceptance of international labor migrations, on the flexibility of prices and wages, and on the political acceptance of *internal devaluations* as a reequilibrating strategy. These conditions are not perfectly met for judging the EU as an optimal currency area in the sense of Mundell (1961), as it was stated by a critical observer like Eichengreen (1990), but also as it was shown long before by many direct observers, suggesting the crucial necessity of strong common fiscal and redistributive policies (Masera 1975).

Productivity convergence among countries is expected to be achieved through the acceptance of *new* macroeconomic tools, on the one hand, and the capacity to take advantage of the microeconomic benefits, on the other hand. Internal devaluation through labor cost decrease was the recent experience of countries like Portugal, Spain, and Greece toward a reequilibrating strategy. The lower transaction costs on currencies were expected to favor mainly weak countries.

The picture on the impact of the common currency on intranational disparities is instead not straightforward. The impact of a macroeconomic policy has been mainly interpreted in the literature as spatially blind. Only recently this erroneous approach has been overcome by serious theoretical reflections suggesting a differentiated spatial impact of macroeconomic policies (Camagni and Capello 2014; Furceri, Mazzola, and Pizzuto 2019). Two types of effects counterbalance a potential divergence expected from an institutional change of this kind (it opens weak areas of weak countries to higher competition). First, the abolition of competition policies obtained through exchange rate variations—generally extremely helpful for the competitiveness of economically strong areas within countries—can particularly hit those regions. Second, enhanced and widened regional development policies are expected to counterbalance the divergent trends (Petrakos 2001; Merler 2016).

The Enlargement to Eastern Countries and Regional Disparities

The last institutional step analyzed from the theoretical point of view is the 2004 enlargement toward the CEECs, followed in 2007 by a much smaller one that involved Bulgaria and Romania. The difference with respect to previous enlargements is that this embraces a high number of countries at the same time, characterized by a decisive lower stage of development, mostly coming from centralized economic systems. For incumbent countries, this opened a large market for final goods and a vast availability of (cheaper) labor forces through FDI. The advantages mentioned for the previous enlargements hold for the new member states. In particular, a clear trend of convergence was foreseen for these countries, through (1) labor cost competitiveness (Casi and Resmini 2014; Popovici and Calin 2015), thanks to the acquisition of knowledge and managerial capacities particularly from FDI (Iammarino and McCann 2013; Casi and Resmini 2017); (2) proximity to Germany, the core of Europe (Petrakos 2001; Pusterla and Resmini 2007)[13]; and (3) the access to structural funds made available to these countries (Resmini 2007).

As in the case of previous enlargements, in this case the short-term effects on intranational disparities are negative. The integration exacerbates divergence within such countries, favoring the core areas and those areas close to the border with the EU. Both these types of regions become the natural location for FDI (Dunford and Smith 2000; Petrakos 2000; Traistaru, Nijkamp, and Resmini 2003). In old member states, intranational divergence is expected, mainly due to the redirection of a large quantity of EU funds from old to new member states, impoverishing the weak regions of the old member states (Heidenreich 2003).

[13] In fact, each enlargement shifted the center of gravity of the EU.

Methodological Approach

The expectations mentioned above on the impact of the different institutional changes have never been empirically tested all at the same time. This is the aim of the present work, requiring a database comparable in time and between different spatial units for all the twenty-eight countries of the EU. This was possible thanks to the publication on the DG REGIO website of data produced by Cambridge Econometrics, which reports real (2005 prices) GDP at the regional level[14] for all countries of the EU since 1980 (although the availability from 1980 only concerns the oldest members and also excludes East Germany). The comparability between different units required some additional work, because real data do not consider differences in purchasing power of the same currency (e.g., the euro) in different countries, differences that reflect different values of each national currency when translated into euros.

For this reason, in this article GDP—on which actual disparities are calculated—was chosen in purchasing power standard (PPS) values, rather than purely monetary ones. GDP PPS was computed for each NUTS2 region for the whole period taking into account national deflators (price deflator GDP: ECU/EUR 2005 = 100), exchange rates (units of national currency per EUR/ECU) and national PPPs (units of national currency per PPS) provided by DG REGIO, following the standard procedure according to the following formula:

$$GDP\ PPS_r = real\ GDP_r\, x\ deflator_c\ x\ exchange\ rate_c / PPS_c \tag{1}$$

where the indexes c and r stand,respectively, for country and region.

The choice of using GDP in PPS is also justified for policy reasons, since the eligibility to European Structural and Investment Funds (ESIFs) assistance is determined based on the relative GDP per capita in PPS in the various EU regions.

Moreover, a synthetic indicator is required for the measurement of disparities, and its choice, among the different possible indicators existing in the literature, is not a trivial one. These measures generally provide a quite consistent picture of the main patterns followed but also have different results because they are differently sensitive to the features of the distribution, including the tails (Ezcurra and Rodríguez-Pose 2009).

Unweighted measures cannot be considered as satisfactory, because the representation they provide is biased by the different sizes of different regions. For instance, the population size of NUTS2 regions in Europe varies from 12.2 million to less than 30 thousand inhabitants (Eurostat 2018).

Among the weighted ones, the one that has been chosen in this context is the Theil index. This index has the advantage of being decomposable in two or more layers (Akita 2003; Capello, Fratesi, and Resmini 2011; Butkus et al. 2018), and this is very important because it allows distinguishing between within-country and between-country disparities.

The decomposable indicator, the Theil index, has been computed on GDP PPS per head according to the following formula, which also shows the decomposition in two components, adding up to the total:

$$T = \sum_r \frac{Y_r}{Y_{EU}} \ln \frac{Y_r/P_r}{Y_{EU}/P_{EU}} = \sum_c \frac{Y_C}{Y_{EU}} \ln \frac{Y_C/P_c}{Y_{EU}/P_{EU}} + \sum_c \frac{Y_C}{Y_{EU}} \sum_{r \subset c} \frac{Y_r}{Y_C} \ln \frac{Y_r/P_r}{Y_C/P_C} \tag{2}$$

where Y stands for GDP PPS, P for population, and the indexes EU, C, and r stand, respectively, for EU, country, and region. As can be clearly seen in equation (2), the total

[14] For this work, data were used at the NUTS2 level.

Theil index (the first element) can be decomposed into a between-country index and a within-country index.[15]

In the sample period of this study, the EU has undergone a series of enlargements, with a different number of countries becoming part of the EU in each one.[16] A longitudinal analysis of regional disparity trends can tell whether theoretical expectations were interpreting reality or not. In fact, we believe in the capacity to obtain interesting results from a trend analysis, since we are dealing with large and deep shocks on societies; launched at precise points in time; hitting rules and conventions, activities, and interests; behavioral codes; and ways of thinking. They are destined to provide opportunities and challenges, to generate huge transformations on economies and societies, sometimes overcoming the effects of other concomitant changes, less pervasive or happening in a smoother way, or sometimes emphasizing their effects. They are accompanied by a host of complementary measures, laws, regulations, and incentives, designed to assure a consistent and orderly implementation of the main institutional change, and to smooth out its potential drawbacks. All this makes the new institutions the most powerful drivers of change and bifurcation in the development trajectories of countries and regions. Of course, effects and adjustments do not happen instantaneously. However, nowadays reactions are becoming faster and faster, and often even take place in advance, following the expectations—more or less rational, nevertheless effective—of the most dynamic actors. This is the logic with which the longitudinal disparity trends of the next sections are analyzed.

The Evolution of Intercountry Disparities in the EU: A Long-Run Convergence

The first analysis deals with the effects of the institutional changes on international disparities. Figure 1 presents the evolution of between-country regional disparities over the period 1980–2015, measured through the first component of the Theil index. Each line represents the level of disparities within a specific group of member countries, with the dashed part representing the time in which the last countries joining that particular group were not yet part of the EU, the continuous line displaying the period in which all countries belonging to the group are officially part of the EU. The position of each line reflects the level of disparities, drastically changing with each widening of the EU, while the slope of each line represents the changes in disparity levels, analyzed at each time of widening and deepening events.

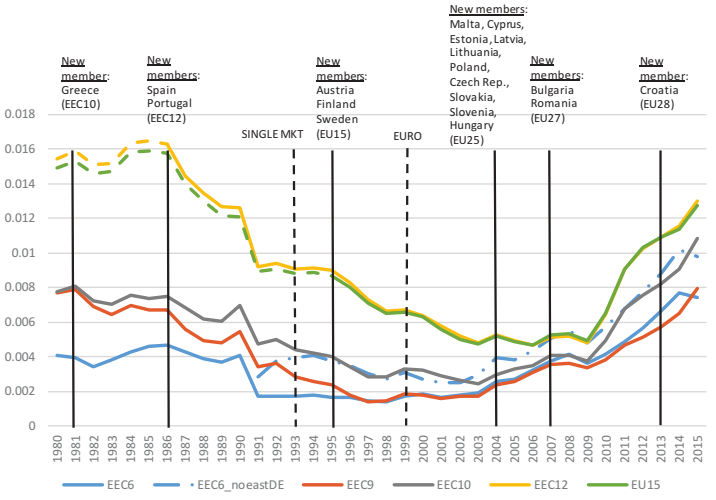
The main message that Figure 1 suggests is that convergence is the norm, whatever the deepening or widening event we take into consideration (Figure 1(b), line EU28). Although each enlargement implied a worsening of between-country disparities in absolute value (upward shift of the curve),[17] what theory mainly suggests in terms of

[15] The weighted coefficient of variation and the mean logarithmic deviation were also considered. The first, in particular, is a very common indicator, especially in official EU Commission documents (European Commission 2017), but has the limit of not being decomposable. In Appendix A it is shown that the Theil index, the weighted coefficient of variation, and the logarithmic deviation provide consistent results. Consequently, the decomposable Theil index is preferred.

[16] 1981: Greece. 1986: Portugal and Spain. 1995: Austria, Finland, and Sweden. 2004: Malta, Cyprus, Estonia, Latvia, Lithuania, Hungary, Czech Republic, Slovak Republic, Poland, and Slovenia. 2007: Bulgaria and Romania. 2013: Croatia. The first two enlargements could not be explicitly taken into account within the present work due to data available only since 1980.

[17] This result is mainly a statistical effect, since, in general, new entrants are poorer with respect to incumbents' average. An exception exists in this regard, represented by the 1995 enlargement, when richer countries—namely, Austria, Finland, and Sweden—entered the EU with a slightly positive effect on between-country disparities.

a) **Evolution of intercountry disparities by groups of countries in the first four enlargements since 1980**



b) **Evolution of intercountry disparities by groups of countries in the last three enlargements**

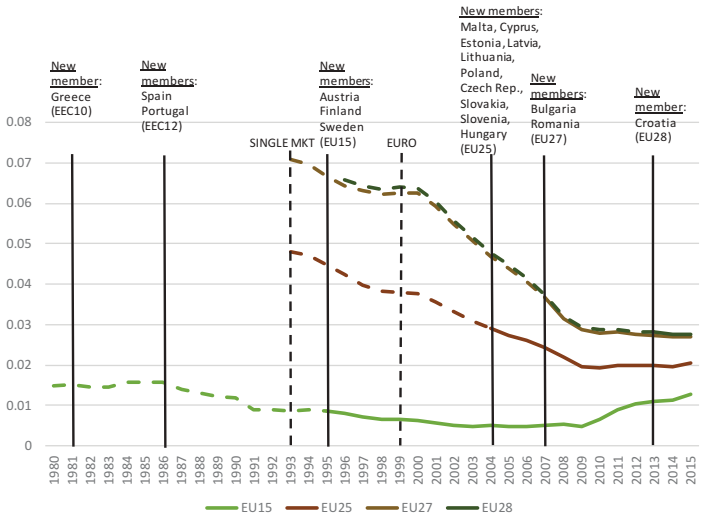


Figure 1. Evolution of intercountry disparities by groups of member countries 1980–2015, Theil index (GDP PPS per head).
Source: Authors' elaboration on Cambridge Econometrics data.

overall decreasing international disparities is confirmed, as shown by the generally negative slope of the lines in Figure 1. In the short term, the scale advantages of a market made larger by the subsequent enlargements show their positive effects.[18]

[18] With the reunification of East and West Germany, disparities among European member countries decreased. This is particularly evident in the evolution of disparities in the group of the six founding members. With the reunification (continuous blue line in Figure 1(a)), disparities decreased in 1991 (the first year for which data on Eastern German regions are available) and remained constant for some

What are not clearly emerging are the anticipatory effects of enlargements on international disparities (dotted lines in Figure 1(a) do not show a negative slope some years before the years of enlargements, that is, before they become continuous lines), with the notable exception of the 2004 accession of CEECs, prepared in advance through precise and effective preaccession measures of cooperative and institutional character (Figure 1(b)).

Concerning the single European market, the expectations on its convergence effects are confirmed. This institutional change brought indeed a quantum jump in political integration and interaction across countries that were already freely exchanging goods (Giannetti 2002). Ex post effects are visible in almost all groups of countries (with the exception of the six founding ones), involving especially the UK, Ireland, and Denmark (between 1993 and 1999 intercountry disparities decrease by 35 percent in the EEC9 group, while they remain substantially stable in the EEC6 group). But the most striking effects were the ex ante ones, generated by the important private investment flows following the launching of the project in 1985, addressed mainly toward capital and financial cities of all EU countries and especially in lagging ones (Ireland and Spain in particular but to a lesser extent also Italy). In Spain and Portugal these anticipatory trends merged with the effects of the recent accession, emphasizing each other (Figure 1(a), EEC12 line). In a worldwide perspective, it is likely that incoming globalization played a role in this convergence process; especially the UK and Ireland were possibly even more advantaged due to lower linguistic barriers.

Empirical evidence corroborated the convergence expectations on the effects of the introduction of the euro, as well. No anticipatory effects on disparities can be observed in relation to this institutional change (in Figure 1(a) no lines show a negative trend in the years before 1999), which influenced investment decisions to a more limited extent with respect to the single market project. However, from 2003, between-country disparities started to increase within the group of EEC10, due to the persistent stalemate of Italy (in spite of its joining the Euro Group with a favorable exchange rate) and a slowdown of the Irish economy due to emerging inflationary processes. The loss of the possibility to utilize competitive policies based on devaluation, previously utilized by countries like Italy and, to a lesser extent, France, could have played a part in these trends, in spite of the widening of the EU budget for cohesion policies.

Concerning the important 2004 enlargement, the empirical evidence is in line with theory (Figure 1(b)), showing new member countries converging in per capita product toward old member countries, thanks to their capacity to attract FDI, coupled with large amounts of structural funds.

These conditions lasted until the crisis, when the general convergence trend was interrupted. CEECs' growth lost momentum while a weakness of Southern countries (Greece, Spain, Portugal, and Italy), and Ireland emerged in full evidence, exacerbated by austerity policies.

Fighting Gravity: Long-Run Intracountry Divergence

The theoretical expectations on intracountry disparities proved generally true, especially for the short-term divergence effects, and because within-country disparities follow a completely different trend with respect to international ones.

years. Such *positive* evolution in disparities is due to Germany becoming poorer and thus moving toward the average of the other countries. The dashed-dotted blue line in Figure 1(a) represents, instead, how disparities would have evolved without the German reunification.

In the group of the six founding countries, contrary to expectations, a true increase in internal disparities happened only after the 2008 crisis, while the establishment of the single market was accompanied by a substantial constancy, reflecting a kind of structural stability achieved in a long period of integration. The true novelty was the German reunification that, beyond the statistical point effect due to the entrance of lagging East German regions (1990–91), determined an increase of internal disparities only up to 1993 due to the closing down of many industrial activities of the former socialist economy (as a comparison between Figure 2(a), (b) witnesses).[19] After that year, internal disparities no longer grew.[20]

A different, diverging, trend in internal interregional disparities following the launching of the single market became evident for the other groups of countries beginning in 1993 (see the divergence of their curves with respect to EEC6 in Figure 2). This fact is linked to the greater capacity of core regions (usually urban areas and capital cities) to take rapid advantage of a wider market and to attract investments in services, and was probably reinforced by the new Information and Communications Technologies (ICTs) paradigm that started in these years, displaying its centripetal forces. At that time, cities became the principal *loci* of higher quality ICTs networks and services, obtaining strategic advantages from new technologies. The result was the cumulative strengthening of core areas with respect to declining regions, as mirrored by increasing divergence.

The establishment of the common currency seems to tell a different story with respect to what theory was suggesting. The theoretical expectations were, in fact, in this case toward potential divergence (Dunford 1994; Boldrin and Canova 2001). However, internal diffusion forces apparently played a relevant role, and intranational disparities registered a clear decrease in all groups of countries at least up to 2006 (Figure 2(a)). The hypothesis that might be proposed refers to the fact that the loss of price-competitive strategies obtained through devaluations probably penalized industrial areas in weak countries more than others. This trend was reinforced by the enhanced and widened regional development policies in favor of weaker regions, and by an initial penetration of the ICTs technological revolution in lagging areas (Barrios, Flores, and Martínez 2019).

The last 2004–7 enlargements were carriers of intranational divergence, as expected. Increasing within-country disparities in this case are the expression of greater integration advantages for advanced (urban) regions, a pattern that was exacerbated by the economic crisis.

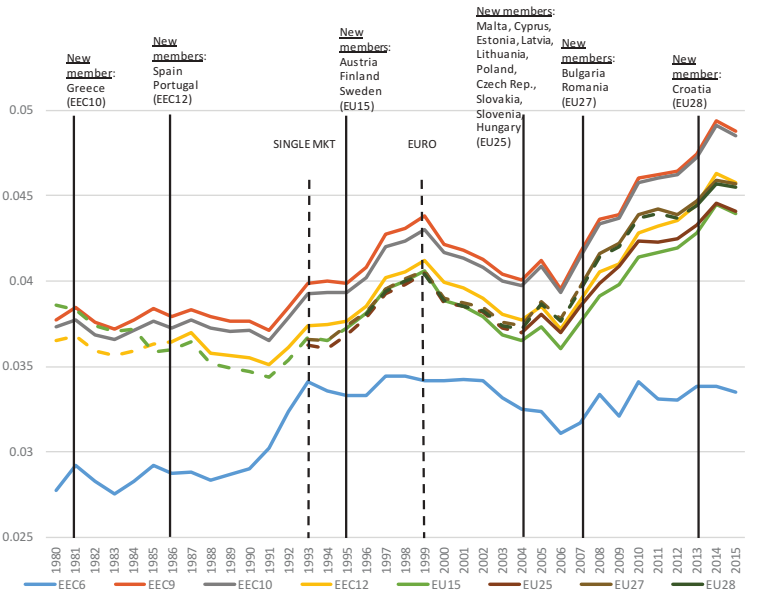
Starting in 2007, overall intranational disparities were raised dramatically in all groups of countries with the consequent rise in social discontent expressed in populistic attitudes, social protests, and emerging anti-European votes in the UK and in the elections for the European Parliament.

European cohesion policies are trying to limit the forces that push toward intracountry regional divergence from multiple directions, from concentrated investment attraction to ICTs adoption: should this be equated to *fighting gravity* (namely, controlling

[19] The light blue line representing the six founding members is much steeper in Figure 2(a) where East German regions are taken into account with respect to Figure 2(b). This is true for all other lines, representing different groups of countries, although not so evident given the lower weight of German regions in the aggregates.

[20] "[T]he united German government invested heavily in eastern infrastructures of all kinds. Physical, administrative, social, educational and scientific infrastructures were torn up at a speed that may in due course be regarded as one of the most remarkable achievements in rebuilding the basics of a modern market economy in just a few years' time." (Paqué 2003, 110).

a) **Evolution of intracountry disparities by groups (including East Germany from 1991)**



b) **Evolution of intracountry disparities by groups (without East Germany from 1991)**

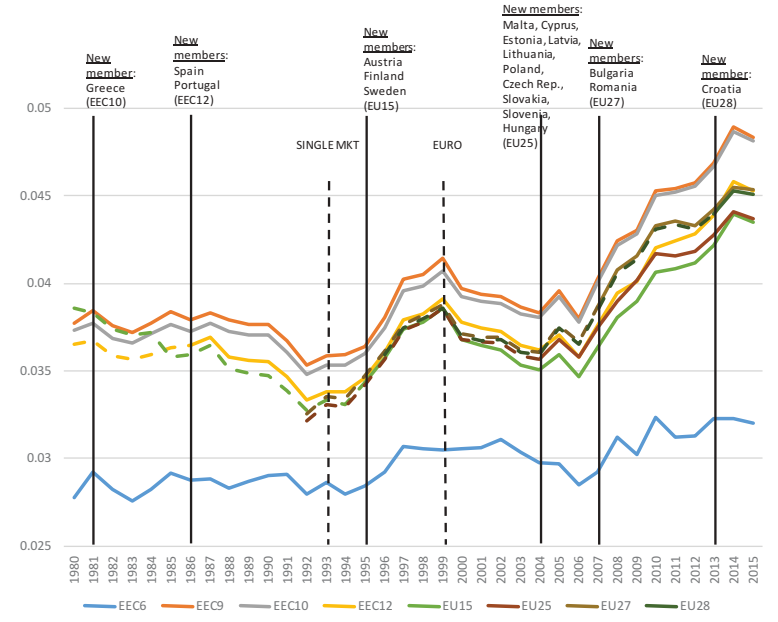


Figure 2. Intracountry disparities by groups of member countries 1980–2015, Theil index, within component (GDP PPS per head).
Source: Authors' elaboration on Cambridge Econometrics data.

some physiological effects of integration) or even *fighting history*, trying to avoid natural cultural, social, and economic differences?

Synchronizing Entry Time in the EU: Intracountry Disparities and Stages of Development

Two issues have not fully been investigated up to now. The first issue is whether the institutional association to the EU systematically causes an increase in intranational disparities; the second one is whether such impact depends on the stage of development of the countries.

Figure 3 addresses the first issue by representing the evolution of within-country disparities and income per head in EU member countries by synchronizing the entry time of each enlargement, and taking four years before and after as a reference for comparison.[21] The descriptive result leads in the direction of claiming that the enlargement has a negative effect on regional disparities that increase in most countries, soon before and after accession. This is especially true for CEECs, which joined the EU already with a very high level of internal disparities, and exacerbated them as the result of the development opportunities given to large cities and core regions with respect to rural or small regions, and of the industrial restructuring process taking place since the fall of the iron curtain. To a lower extent, this trend is shared by Western countries, which also register an increase in disparities after joining the EU, even if to a limited extent. A peculiar case is Greece, which did not experience a significant rise in its internal disparities but became even relatively poorer (Figure 3). Greece represents, in general, a very peculiar case, since at the time of its entry, it did not share any borders with other EU countries, did not get preaccession funds, and did not have a strong industrial know-how.[22]

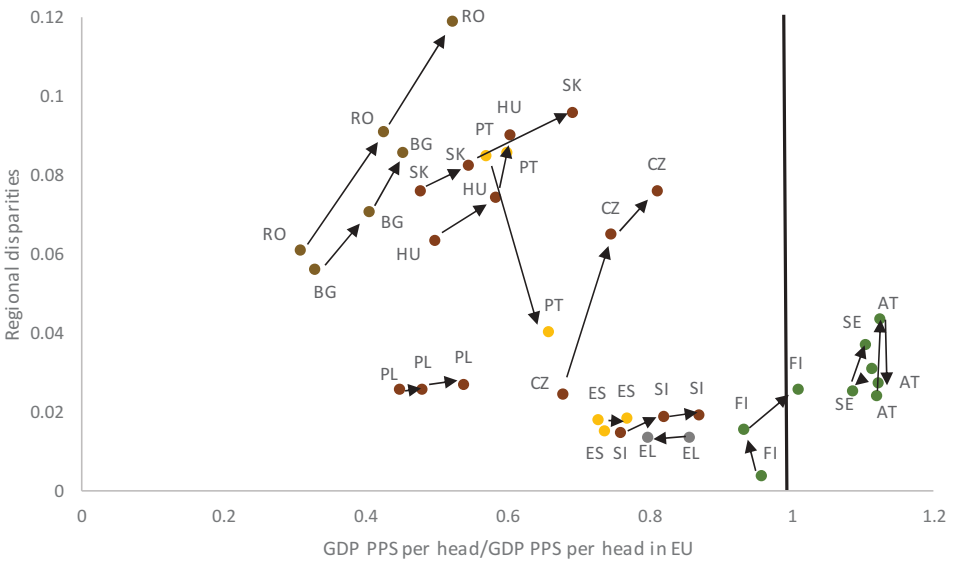


Figure 3. Regional disparities and relative income per head by country at the EU entry year and four years before and after.
Source: Authors' elaboration on Cambridge Econometrics data.

[21] Four years have been chosen as a reasonable time to grasp the short-term effects of entering the EU; it is short enough to be available for those member countries that entered very recently and long enough to capture some tendencies.

[22] Campos, Coricelli, and Moretti (2019) even find negative effects from EU membership in the case of Greece.

Overall, the increase of regional disparities represents a continuous trend already registered four years before the accession takes place, witnessing the presence of some anticipatory effects. In some countries, like Hungary, this pattern is reinforced the moment the country joins the EU; in most other cases, however, this is a constant trend. A very important normative message comes out of Figure 3. Disparities in Eastern countries are very high, much higher than in all other countries, and, more importantly, they grow much more than elsewhere. This trend calls for an urgent intervention in order to avoid within-country disparities in the Eastern countries to remain permanently different from those in Western countries.

The second issue is addressed in Figure 4(a)–(c), displayed with the aim of linking disparities with level of development. The increase of disparities, in fact, ought to take place in the catching-up phase of less developed national economies based on most modern and advanced areas, while advanced economies are expected to be closer to a frontier of decreasing returns on (spatially concentrated) investments, as suggested by Williamson's (1965) theory, following the notorious inverted U-shaped trajectory (Ezcurra and Rapún 2006; Barrios and Strobl 2009; Lessmann 2014). However, technological progress and the evolution of knowledge are factors that may give advanced regions a chance for raising returns on capital, and consequently for continuing to attract capital and labor from weaker regions and to concentrate public investments in advanced infrastructure (e.g., hub airports, high-speed trains). Consequently, in the advanced regions, the frontier of decreasing returns on investments can be pushed toward higher levels of income.[23]

In the case of CEECs, the pattern followed by the Theil index seems to confirm the Williamson provision that regional disparities increase during the early phases of country development and decrease after a certain level of income is reached (Figure 4(a)). The empirical evidence seems to provide two insights: on the one hand, the tendency to increase disparities seems to take place for all countries independently from the initial level of disparities; on the other hand, the turning point seems to be reached for different levels of per capita income. It in fact takes place for less than 12,000 euro PPS in Bulgaria, up to about 18,000 euro PPS in Slovakia.[24]

One of the CEECs, the Czech Republic, shows a curve with two peaks that, as we shall see, is typical of richer countries. Northern countries, in fact, show two inverted U-shaped curves, more pronounced in countries such as the UK, Sweden, Austria, and less pronounced in France, Finland, and Germany. The curves show that the country has already gone through two cycles (Figure 4(b)). After a period of fall, in the presence of technological progress and social change, regional disparities increase again. Within this group, two countries show a peculiar pattern, which suggests an increasingly concentrated growth before, during, and after the crisis, namely, France and the UK. Not surprisingly, these are the same countries in which populistic movements gained momentum in recent years.

Although the pattern is much less evident, Southern countries show double inverted U-shaped curves similar to those in the Northern countries (Figure 4(c)).

[23] A debate exists in the literature on the actual presence of this inverted U curve, or rather the presence of a J curve (Petrakos, Kallioras, and Anagnostou 2011; Monastiriotis 2014; Artelaris and Petrakos 2016).

[24] In the previous period before the one considered in this work, namely, from the creation of the European Economic Community in 1957 to 1980, the Williamson curve proved right. Up to 1970 disparities increased, especially in weaker countries, like Italy and Belgium, but then decreased: large cities were left by industry, old industrial areas were hit by the crisis of their specialization sectors (iron and steel, shipbuilding, textiles) and new intermediate regions emerged (the *Third Italy*, that is the northeastern and central regions of the country, Flanders, Southern France, Southern Germany) (see Camagni and Cappellin 1985; Camagni 1995).

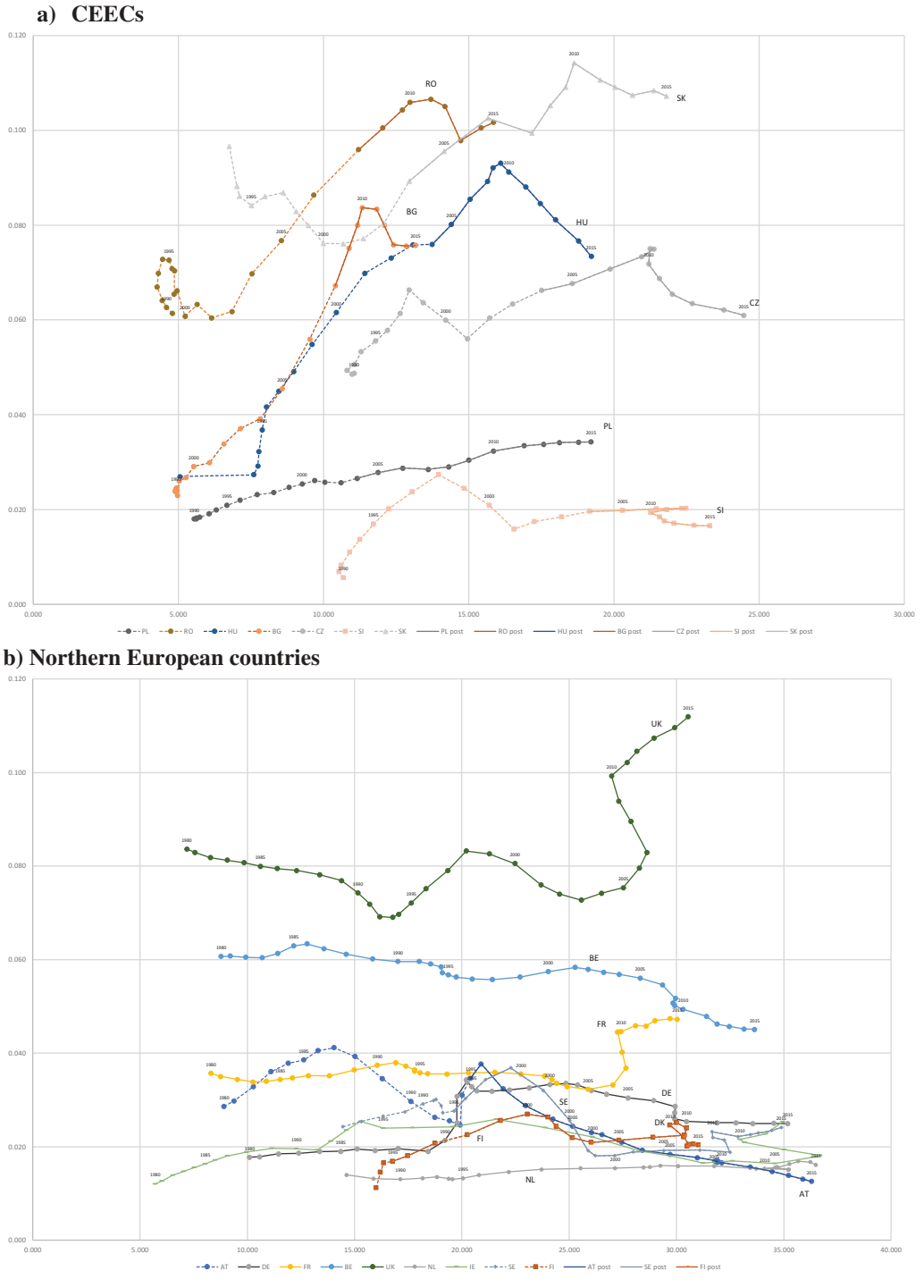


Figure 4. (Continued on next page).

c) Southern European countries

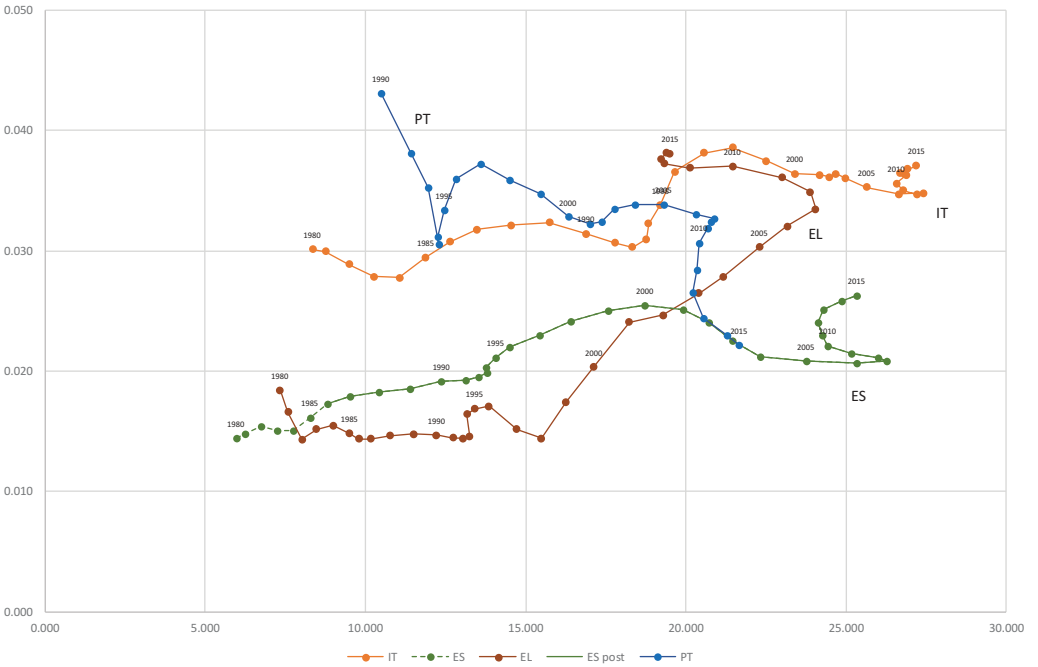


Figure 4. Relationship between regional disparities and income per head: Williamson's curve legend: dotted lines refer to years in which a specific country was not yet part of the European Union. Source: Authors' elaboration on Cambridge Econometrics data.

In this case, the turning point is achieved at lower levels of disparities. In particular, Italy and Spain at the start of the 2008 crisis moved toward an increase in disparities, similar to the pattern shown by the UK and France, but in the presence of a reduction in GDP for many years (Ubago Martínez et al. 2019; Graziano et al. 2019).

The moment in time at which disparities start a new cycle takes place in different years in different countries. This is most likely because the new technological paradigm does not spread everywhere at the same moment. Countries and regions grasp the advantages of a new paradigm at different moments in time, according to the time span with which they learn how to exploit the new technologies. Within a conceptual regularity like the one presented by the Williamson curve, specificities emerge. The relationship between disparities and per capita income level, in most cases, is not affected by the accession of countries in the EU. The dashed lines in Figure 4, representing the time in which countries are not yet EU members, show the same pattern as the solid lines, capturing the relationship in periods of membership. This is particularly evident in CEECs, where both the years before and after the accession are characterized by increasing disparities, and the same seems to be true in Southern and Northern countries, although less clearly and with the exception of Portugal.[25]

[25] As also highlighted by Petrakos, Rodríguez-Pose, and Rovolis (2005), the sharp fall in the Theil index in the case of Portugal depends on the fluctuations in the GDP in the region of Alentejo during the 1980s. As a consequence, these data should be viewed with caution.

Causal Link between Institutional Changes and Regional Disparities: An Event Analysis Approach

The discussion provided in the previous sections can find additional validation in an econometric analysis aimed at establishing a causal link between the institutional changes considered in the whole work and the evolution of regional disparities across the EU. In particular, an event analysis approach was applied (see Karafiath 1988; Jasmand and Maenning 2008; Monastiriotis, Kallioras, and Petrakos 2017), treating the different institutional changes (enlargement/accession, advent of the single market, and introduction of the common currency) as unique events that may affect the growth of regional disparities and controlling for other elements that could have potentially influenced this relationship.

In greater detail, a pooled ordinary least squares model was estimated, where the yearly variation in intranational disparities was made dependent on dummies associated with the specific *event* (institutional change), and on control variables, as follows:'

1. $\Delta Theil_{ct} = \alpha + \beta_1 preaccess_{ct} + \beta_2 postaccess_{ct} + \beta_3 long\ term\ access_{ct} + \beta_4 \Delta\ sh\ emp\ agr_{ct} + \beta_5 \Delta\ sh\ emp\ ind_{ct} + \beta_6 \Delta\ sh\ emp\ fbs_{ct} + \beta_7 \Delta\ investments_{ct} + \beta_8 \Delta\ GDP\ per\ capita_{ct} + \beta_9\ GDP\ per\ capita_{ct} + \beta_{10} country + \beta_{11} time + \varepsilon_{ct}$

2. $\Delta Theil_{ct} = \alpha + \beta_1 presingle\ mkt_{ct} + \beta_2 postsingle\ mkt_{ct} + \beta_3 long\ term\ single\ mkt_{ct} + \beta_4 \Delta\ sh\ emp\ agr_{ct} + \beta_5 \Delta\ sh\ emp\ ind_{ct} + \beta_6 \Delta\ sh\ emp\ fbs_{ct} + \beta_7 \Delta\ investments_{ct} + \beta_8 \Delta\ GDP\ per\ capita_{ct} + \beta_9\ GDP\ per\ capita_{ct} + \beta_{10} country + \varepsilon_{ct}$

3. $\Delta Theil_{ct} = \alpha + \beta_1 pre\text{-}euro_{ct} + \beta_2 post\text{-}euro_{ct} + \beta_3 long\ term\ euro_{ct} + \beta_4 \Delta\ sh\ emp\ agr_{ct} + \beta_5 \Delta\ sh\ emp\ ind_{ct} + \beta_6 \Delta\ sh\ emp\ fbs_{ct} + \beta_7 \Delta\ investments_{ct} + \beta_8 \Delta\ GDP\ per\ capita_{ct} + \beta_9\ GDP\ per\ capita_{ct} + \beta_{10} country + \varepsilon_{ct}$

where the units of analysis are the EU countries, *c*, with more than 2 NUTS2 regions according to the 2013 nomenclature (nineteen countries)[26] during the period 1980–2015 (*t*).[27] The dependent variable ($\Delta Theil_{ct}$) is the yearly growth rate of the national Theil index, representing the evolution of intracountry disparities.

In the first specification, the events taken into consideration are the different enlargements, and the aim of the analysis is to capture the trend of disparities in three moments in time: (1) in the preaccession period (four years before each specific enlargement), with a dummy variable equal to one for the four years before the accession to the EU ($preaccess_{ct}$), (2) in the postaccession period, with a dummy variable equal to one for the year of accession and the four following years ($postaccess_{ct}$); and (3) in the following most recent years after the postaccession period, with a dummy variable equal to one, from the fifth year after the accession onward ($long\text{-}term\ access_{ct}$). The descriptive analysis suggests that the entrance into the EU has a negative effect on intraregional disparities, since the advantages are mainly caught by the richest areas of countries. The dummies are country specific, since not all members joined the EU at the same time. This is also the reason why time-fixed effects are added to country-fixed effects in this first specification.

The second and third specifications are meant, instead, to catch the effects of the advent of the single market and of the introduction of the common currency on national

[26] Excluded countries are Cyprus, Estonia, Latvia, Lithuania, Luxembourg, and Malta. In addition, Croatia, Ireland, and Slovenia were dropped, since by having only 2 NUTS2 regions according to the 2013 nomenclature, their Theil index may be unreliable.

[27] In fact, CEECs enter the database only at the beginning of the 1990s so that the overall sample is unbalanced in terms of number of observations over time.

disparities, respectively. In this case, the descriptive analysis seems to suggest that participation in the common currency increases disparities only before the event and not after. The dummy variables enter the regressions according to the same logic explained above, therefore taking into account the four years before the events, the four years after the events, and the following period. The periods are therefore 1989–92, 1993–97, and 1998 onward for the advent of the single market, and 1995–98, 1999–2003, and 2004 onward for the introduction of the common currency. Since in these cases the periods are the same for all the countries, time-fixed effects could not be included in the regressions.

Controls are common to all specifications, and are inserted in order to capture (1) the evolution of the stages of development, through the variation of the employment structure, and in particular in the shares of employment in agriculture, industry, and financial and business services (Δ *sh emp agr*$_{ct}$, Δ *sh emp ind*$_{ct}$, Δ *sh emp fbs*$_{ct}$). An increase in the share of employment of low value-added sectors is expected to be negatively linked to regional disparities; (2) the dynamism of the economy, through the growth rate of gross fixed capital formation over GVA (Δ *investments*$_{ct}$), expected to be positively associated to regional disparities given the concentration of investments in strong areas of the countries; (3) the growth of the national wealth (Δ *GDP per capita*$_{ct}$), expected to be negatively associated to regional disparities, since most countries have already reached a certain level of wealth; (4) the wealth of the initial period, to control for other elements than the sectoral dimension that could characterize the stage of development of a country (*GDP per capita*$_{ct}$).[28]

The results are displayed in Table 2 and are overall consistent with our previous analyses. Regression (1) shows how internal disparities increase significantly just before the accession and keep on increasing just after it, in line with what is displayed by Figure 3. On the other hand, long-run membership does not seem to be related to a growth in disparities. The result is similar for the advent of the single market (column 2 in Table 2; see also Figure 2), although in this case the positive contribution seems to persist also in the long run. The specification reported in column (3) of Table 2 tells instead a different story about the introduction of the common currency: after increasing disparities just before the event, there is no sign of increasing disparities after it. This confirms our previous analysis and in fact contradicts the theoretical expectations. As mentioned before, some mitigating forces (e.g., regional development policies and penetration of the ICT paradigm) were probably quite effective in avoiding a sharp growth in disparities.

The *R*-squared for the different specifications are low. However, since the aim of the exercise is rather to highlight the causal link between particular institutional changes and the evolution of disparities than to explain regional disparities in their entirety, the significance of the dummy coefficients and the *F*-tests guarantee the significance of the results.

The three models already focus on different periods and provide different results. Nevertheless, an additional robustness check was run on the results. To be sure that the outcomes do not just capture a period effect, following Monastiriotis, Kallioras, and Petrakos (2017), a *placebo* regression was performed (column 4 in Table 2) with a set of dummies corresponding to three different sequential and casual periods: 1997–2000, 2001–5, and 2006 onward. The periods were chosen to not overlap with those previously used and to include all the countries in the sample. As clearly shown in Table 2, the results are very dissimilar. The three coefficients of interest often have the opposite sign and are, in any case, not statistically significant, implying that the main

[28] All the data were retrieved from Cambridge Econometrics.

Table 2

Econometric Results of the Event Analysis

	EU Enlargement Widening (1)	Single Market Deepening (2)	Common Currency Deepening (3)	Placebo (4)
Preaccess	0.109**			
	(0.050)			
Postaccess	0.072**			
	(0.037)			
Long-term access	0.030			
	(0.028)			
Pre–single market		0.061*		
		(0.036)		
Post–single market		0.069**		
		(0.030)		
Long-term single market		0.080**		
		(0.033)		
Pre-euro			0.050**	
			(0.026)	
Post-euro			0.022	
			(0.026)	
Long-term euro			0.048	
			(0.034)	
Placebo (1997–2000)				0.001
				(0.021)
Placebo (2001–5)				−0.010
				(0.024)
Placebo (2006–15)				−0.027
				(0.031)
Δ share employment agriculture	−1.163	−1.422	−1.815*	−1.644
	(1.160)	(1.028)	(1.030)	(1.032)
Δ share employment industry	−1.802	−1.831	−2.589*	−1.907
	(1.846)	(1.539)	(1.614)	(1.568)
Δ share emp financial & business services	−3.093	−2.246	−3.002	−2.757
	(2.589)	(2.440)	(2.418)	(2.393)
Δ investments	0.275	0.150	0.144	0.187
	(0.496)	(0.456)	(0.472)	(0.467)
Δ GDP per capita	−0.645**	−0.573***	−0.622***	−0.661***
	(0.318)	(0.205)	(0.211)	(0.206)
GDP per capita (level)	−0.006*	−0.004**	−0.003	0.001
	(0.004)	(0.002)	(0.002)	(0.002)
Constant	0.052	0.046	0.059	0.004
	(0.074)	(0.056)	(0.064)	(0.057)
Country fixed effects (FE)	YES	YES	YES	YES
Time FE	YES			
No. of obs.	594	594	594	594
Prob > *F*	0.003	0.003	0.001	0.008
R^2	0.154	0.070	0.065	0.058

Note: Robust standard errors in parentheses. Significance levels as follows: ***1%, **5%, *10%.

regression results are linked to the events and not to some random time trends. In addition, the *F*-test does not provide a significant value and the *R*-squared is lower than in the previous regressions. This gives us confidence that our results are indeed substantially linked to the institutional changes we are interested in.

Conclusions

Starting from a complete and consistent database on GDP in PPPs for all countries and regions in the EU since 1980, this article analyzes the entire process of relative growth of countries and regions from the point of view of regional disparity trends. The assumption, to be confirmed, is that the main drivers of transformation were the

successive phases of the construction of the EU and its successive enlargements toward new countries. In fact, the milestone dates indicating the starting point of these phases brought in some synchronized discontinuities and powerful new institutions, capable of revolutionizing economic and social behavior and political attitudes. The question that arises is consequently, for the benefit of whom?

The answer to this question is sought on two related dimensions: the theoretical one, examining analyses, expectations, and ex ante predictions of main scholars, and the empirical one, of ex post evidence, in search of any consistency. The question looks crucial, as the main theories and scientific debates on customs and monetary unions mainly concerned overall advantages and necessary conditions, and implicitly referred to the case of countries. However, the condition of countries and regions in unions is very different. The theory of international trade assumes no international migration and benefits to all partners from specialization; in the case of regions, migrations are the rule, and adjustments happen thanks to them, with the risk of desertification in the extreme case of internal, inaccessible areas. Beyond that, regions lack some powerful adjustment mechanisms against recession, namely, monetary policies and control on exchange rates, and a large part of their wage levels come from national bargaining processes: different from countries, they compete on an *absolute advantage* regime (Camagni 2002).

Therefore, in discussing regional disparities, and consequently the differential in regional growth rates, it looks crucial to separate the part that depends on the behavior of countries—bearing a strong influence on all internal regions—and the part depending on interregional differences proper, internal to each country.

The work showed that, within the EU, the long-term, theoretically expected, convergence processes—leading to higher stages of development through information diffusion, integration of local cultures and know-how, strong imitation processes in economic activities and in lifestyles—still have to be accomplished.

In fact, the subsequent enlargements acted positively on the possibilities of weaker national economies to reinforce, modernize, and integrate with more advanced economies, reducing international disparities. However, at the same time they triggered (and exacerbated) intranational disparities, since strong urban areas were the fastest to take advantage of the integration process, widening the gap with weaker regions. All these findings are reinforced by an econometric event analysis, demonstrating a cause–effect link of such phenomena.

Concerning intranational disparities, the expectations about a reversal of the increasing disparity trend coming from the well-known Williamson curve are widely confirmed, with one interesting complement: the inverted U shaped curve of internal disparities can be replicated from the beginning (being shifted to the right of the time axis) as a consequence of a change in the techno-economic paradigm, requiring brand new competitive assets. In this case, disparities can start increasing again, even if they were already lowering during the previous *régime*. Another element that proves relevant in the generation of internal disparities after the EU accession is the average stage of development of the single country, lagging countries being much more exposed to an increase in regional disequilibria.

Counteracting such tendencies, mainly taking place internally to each country, remains a political and policy must. In fact, it could look like *fighting gravity* or *fighting history*, since disparities and disequilibria continuously reappear in different forms and spatial levels, transmitting a sense of unescapable normality. Cohesion policies have certainly played a role in moving in this direction and, in spite of the difficulty of measuring their net effects, it is reasonable to think that the situation would have been worse in case of their absence.

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Appendix A: Robustness of the Analysis to Different Disparity Indicators

In the article, the choice of the Theil index is based on two advantages that are not shared by other indicators: it is decomposable in national and regional components, and it is weighted.[29]

In this appendix, robustness checks of the results are presented by comparing the ones obtained with the Theil index with those of two other indicators, each of them having one of the two properties and both being diffused in the literature.[30]

Figure A1 reports the comparison of the Theil index with the weighted coefficient of variation (WCV). The latter is a commonly used measure in the policy literature (see EU Commission 2017, 4). To obtain long-run comparability, the total Theil and the WCV are calculated only for those regions whose data are available in the database since 1980, excluding therefore the newest members and East Germany. As can be observed in the figure, the two indicators coincide very well.

The mean log deviation (MLG), calculated according to the formulation by Brewer and Wren-Lewis (2016, 295), is a specific case of the generalized entropy index, and shares with the Theil index the advantage of being decomposable. The paths of weighted MLG are identical to those obtained through the Theil index.



Figure A1. Comparing the evolution of regional disparities in Europe in terms of GDP PPS per capita —Theil index (left axis) vs weighted coefficient of variation (right axis).
Source: Authors' elaborations on Cambridge Econometrics data.

[29] Smaller regions weight less and generally have more *noisy data*, due to smaller sample sizes, which introduce biases when analyzing changes in disparities.

[30] See also Ezcurra, Pascual, and Rapún (2007b).